

Report: HHT–BAO Locking Analysis

Executive Summary

We investigated Hilbert–Huang Transform (HHT) decomposition of supernova (SN) Ia residuals to test for possible scale-locking with baryon acoustic oscillation (BAO) effective scales.

By extracting intrinsic mode functions (IMFs) and comparing their instantaneous-frequency ridges with BAO-derived wavenumbers, we tested baseline,

Noether-constrained, and $\Lambda(z)$ -coupled models.

Results demonstrate a robust IMF 2 correlation across all models and smoothing choices, and an emergent IMF 3 lock only under $\Lambda(z)$ coupling.

Background

This investigation is embedded in the PLamb cosmological framework, which explores variable-mass and variable- c dynamics relative to standard Friedmann–Robertson–Walker (FRW) cosmology.

The analysis leverages supernova (Pantheon+SH0ES) data and compares to BAO distance scales.

Models tested:

- FRPBH baseline (variable mass, variable c , PBH-driven acceleration).
- Noether symmetry (conserve_time: freezes accretion terms).
- $\Lambda(z)$ coupled OU process with gamma coupling (fluctuation-induced).

Acronyms

- HHT – Hilbert–Huang Transform.
- IMF – Intrinsic Mode Function (oscillatory component from EMD/EEMD).
- BAO – Baryon Acoustic Oscillation.
- k_{eff} – effective BAO wavenumber from DM/ r_d measurements.
- SN – Supernova Ia distance modulus data (Pantheon+SH0ES).

Key Equations

HHT decomposition: $\text{Residual}(z) = \sum \text{IMF}_i(z) + \text{trend}(z)$.

IMF energy: $E_i = \sum \text{IMF}_i(z)^2$.

BAO effective scale: $k_{\text{eff}}(z) = 2\pi / (D_M(z)/r_d)$.

Method

SN residuals were detrended and placed on a uniform $\log(1+z)$ grid with tapering. Empirical Mode Decomposition (EMD/EEMD) extracted IMFs, and Hilbert transforms provided instantaneous frequencies.

IMF ridges were compared to BAO scales using correlation (Pearson, Spearman) and permutation-based significance testing.

FDR correction identified robust IMFs.

Smoothings of IMFs 5, 7, 9, 11 were tested to confirm stability.

Results

Findings by model:

- Baseline FRPBH: IMF 2 significant across all smoothings ($r \approx 0.14$, $p \approx 5e-4$).
- Noether conserve_time: IMF 2 strengthened ($r \approx 0.22$, $p \approx 5e-4$).
- $\Lambda(z)$ coupling: IMF 2 weaker but still significant ($r \approx 0.09$, $p \approx 8e-3$).
 - Additionally, IMF 3 becomes robustly significant ($r \approx 0.14$, $p \approx 5e-4$).
- Higher IMFs (4–6) showed no consistent correlations ($p > 0.1$).
- Robustness analysis confirmed IMF 2's persistence across all smoothing kernels.

Interpretation

IMF 2 represents a universal locking to BAO transverse scale.

The symmetry constraint (conserve_time) suppresses extraneous structure and enhances the IMF 2 lock.

$\Lambda(z)$ coupling shifts coherence to IMF 3, suggesting an additional frequency band induced by fluctuations.

Conclusions and Next Steps

Evidence indicates HHT-extracted IMFs track BAO scales.

IMF 2 is a stable universal feature; IMF 3 emerges under $\Lambda(z)$ coupling.

Next steps include mapping IMF median frequencies to physical lengths, testing radial (DH/rd) versus transverse (DM/rd) BAO dependence, and jackknife resampling of SN data to confirm robustness.